Participant	F_{CE}^{max} [N]	L_{CE}^{opt} [cm]	L_{SEE}^{0} [cm]	**RMSD [%]**
1	4400	8.6	17	7.3
2	5100	9.9	15	6.9
3	5600	8.7	17	5.8
4	6800	8.6	16	4.1
5	9200	7.4	18	4.7
6	8500	8.8	16	2.9
AVG±SD	6600 ± 1700	8.7 ± 0.7	17 ± 1	5.3 ± 1.6

F_{CE}^{max}, L_{CE}^{opt} and L_{SEE}^{0} were estimated from the isometric measurements for each participant individually. The root mean square difference (RMSD) was normalised by the individual maximal predicted knee joint moment, which is the product of F_{CE}^{max} and the moment arm of *m. quadriceps femoris*.

	Condition								
	0.20	**0.25**	**0.30**	**0.35C**	**0.40**	**0.35A**	**0.35B**	**0.35D**	**0.35E**
W_{ext}^{+}	102.3 ± 2.5	102.2 ± 2.0	103.4 ± 3.2	103.3 ± 4.3	104.2 ± 3.5	104.5 ± 4.0	102.5 ± 3.6	101.9 ± 4.2	100.5 ± 2.7
W_{flex}^{+}	1.3 ± 1.2	0.9 ± 0.9	1.1 ± 1.2	0.9 ± 0.7	1.4 ± 1.0	1.6 ± 1.5	1.7 ± 1.2	2.6 ± 1.9	3.3 ± 1.8
W_{ext}^{-}	-0.0 ± 0.1	-0.0 ± 0.0	-0.0 ± 0.0	-0.0 ± 0.0	-0.0 ± 0.0	-0.0 ± 0.0	-0.0 ± 0.0	-0.0 ± 0.0	-0.1 ± 0.4
W_{flex}^{-}	-3.7 ± 2.2	-3.1 ± 2.0	-4.6 ± 3.3	-4.3 ± 4.1	-5.6 ± 3.4	-6.0 ± 3.6	-4.3 ± 3.0	-4.7 ± 4.4	-3.8 ± 2.0

Positive (+) and negative (-) mechanical work during knee extension (ext) and flexion (flex) were calculated by integrating the instantaneous mechanical power over time, separately for the periods when the instantaneous power was positive (positive work) and negative (negative work) respectively.

	Condition								
	0.20	**0.25**	**0.30**	**0.35C**	**0.40**	**0.35A**	**0.35B**	**0.35D**	**0.35E**
									Movement characteristics
CF [Hz]	0.44	0.59	0.75	0.93	1.13	0.79	0.83	1.13	1.35
FTS [ ]	0.35	0.40	0.45	0.51	0.58	0.72	0.61	0.46	0.48
$\boldsymbol{T_{ext}}$ **[s]**	0.79	0.67	0.60	0.55	0.51	0.91	0.73	0.41	0.36
$\boldsymbol{T_{flex}}$ **[s]**	1.46	1.02	0.73	0.53	0.38	0.36	0.48	0.48	0.38
									AMPO
Participant 1 [s^{-1}]	0.17	0.23	0.30	0.36	0.44	0.34	0.36	0.39	0.40
Participant 2 [s^{-1}]	0.17	0.23	0.26	0.32	0.37	0.32	0.30	0.32	0.35
Participant 3 [s^{-1}]	0.16	0.19	0.23	0.29	0.36	0.27	0.29	0.32	0.34
Participant 4 [s^{-1}]	0.17	0.21	0.27	0.30	0.35	0.30	0.29	0.30	0.35
Participant 5 [s^{-1}]	0.14	0.17	0.22	0.25	0.30	0.26	0.26	0.26	0.26
Participant 6 [s^{-1}]	0.17	0.20	0.24	0.29	0.33	0.31	0.30	0.30	0.32
Average ± SD [s^{-1}]	0.16 ± 0.01	0.21 ± 0.02	0.25 ± 0.03	0.30 ± 0.03	0.36 ± 0.04	0.30 ± 0.03	0.30 ± 0.03	0.32 ± 0.04	0.34 ± 0.04
Predicted [s^{-1}]	0.20	0.25	0.30	0.35	0.40	0.35	0.35	0.35	0.35

CF denotes cycle frequency, FTS denotes fraction of cycle time spent shortening, T_{ext} and T_{flex} denote the knee joint extension and flexion time per cycle. Measured AMPO was normalised to the product of the participant's individually estimated F_{CE}^{max} and the generic model parameter value of L_{CE}^{opt}, while predicted AMPO was normalised to the product of model parameters F_{CE}^{max} and L_{CE}^{opt} (see Methods).

Description	**Symbol**	**Unit**	**Value**	**Reference**
				Musculoskeletal geometry
MTC length at ϕ_{knee} = 0	A_0	cm	21.3	**?**
Linear coefficient of MTC-ϕ_{knee} relation	A_1	cm/rad	4.20	**?**
Hill constant	a^{rel}	-	0.41	**?**
				Contraction dynamics
Hill constant	b^{rel}	1/s	5.20	**?**
Maximum isometric CE force	F_{CE}^{max}	N	5,250	**?**
PEE stiffness scaling parameter	k_{PEE}	N/m¡sup¿2¡/sup¿	1.19×10^7	**?**
SEE stiffness scaling parameter	k_{SEE}	N/m¡sup¿2¡/sup¿	1.28×10^8	**?**
CE optimum length	L_{CE}^{opt}	cm	9.30	**?**
PEE slack length	L_{PEE}^{0}	cm	13.0	**?**
SEE slack length	L_{SEE}^{0}	cm	16.0	**?**
				Excitation dynamics
Determines CE force-length relation width	w	-	0.56	**?**
Determines steepness γ-q relation	a_{act}	-	-4.59	**?**
Determines pCa^{2+} level at which q = 0.5	$b_{act,1}$	-	5.17	**?**
Determines pCa^{2+} level at which q = 0.5	$b_{act,2}$	-	1.08	**?**
Determines pCa^{2+} level at which q = 0.5	$b_{act,3}$	-	-0.19	**?**
Relates γ to pCa^{2+}	kCa	mol/L	8.00×10^{-6}	**?**
Activation time constant	τ_{act}	ms	88.9	**?**
De-activation time constant	τ_{deact}	ms	88.9	**?**

Parameter values are depicted only for those key to predict the maximally attainable AMPO.

Table 1: Your table caption

Label	Description	0.40	0.35C
WposExt	Work positive extension	12.1 ± 1.2	11.5 ± 1.1
WposFlx	Work positive flexion	8.3 ± 0.9	7.9 ± 0.8 .

Table 2: Auto-stretching table

Label	0.40	0.35C	0.30	0.25	0.20	0.35A	0.35B	0.35D	0.35E
Label	0.40	0.35C	0.30	0.25	0.20	0.35A	0.35B	0.35D	0.35
WposExt	12.1 ± 1.2	13.2 ± 1.1	13.2 ± 1.1	13.2 ± 1.1	13.2 ± 1.1	13.2 ± 1.1	13.2 ± 1.1	13.2 ± 1.1	13.2 ± 1.1